

A Prompt-Engineered LLM Framework for Extracting ESA Narratives with Narrative Policy Framework-Based Coding.

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Abstract

Write your abstract here.

CCS Concepts

• Information systems → Question answering.

Keywords

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Check out how the papers of Chris comes in

1 Introduction

Rapid biodiversity loss and growing extinction risk have made public understanding of conservation policy a critical component of environmental governance. In the United States, the Endangered Species Act (ESA) was passed by Congress in 1973 as the primary federal instrument for responding to this crisis [1]. The ESA is widely regarded as one of the most powerful and controversial conservation laws, designed to prevent extinction by listing species as threatened or endangered, protecting their habitats, and directing federal agencies to avoid actions that endanger listed species [2]. Yet despite its scientific and legal foundations, the ESA is rarely encountered by the public through direct engagement. Instead, most people come to understand the law through the stories told about it — stories about which species are at risk, who is responsible for that risk, and what should be done to address it. [3]. Wildlife policy conflicts involving species such as wolves and grizzly bears show this clearly: decisions about wildlife management often reflect deeper cultural and political differences, and the disagreement is usually about more than just facts [? ?]. In Idaho, debates over wolf recovery show this especially well, as the conflict has involved not only

ecological questions but also politics, identity, and competing views about management [?]. These conflicts show that conservation issues are not only about science, but also about values, authority, and control. In that sense, environmental policy debates rarely turn on facts alone: the ESA is not only a legal and scientific framework but also a politically contested issue that is often framed through narratives about which species are at risk, who is responsible for that risk, and what should be done to address it rather than through policy facts alone.

This narrative quality of ESA debate is not incidental to the policy process, it is part of how the policy process operates. Public policy issues are often communicated through competing stories about who is responsible, what the problem is, and what should be done. This indirect relationship between the public and environmental policy elevates the role of media from a passive information channel to an active political actor. Media institutions translate complex scientific and legal information into accessible accounts, selectively emphasizing certain issues, actors, and interpretations while downplaying others. Through agenda-setting, the frequency and prominence of coverage shape which ESA issues and species are perceived as important, bringing some policy questions to the forefront while leaving others largely invisible [4]. Through framing, media narratives organize these issues into coherent interpretations by defining problems, assigning causal responsibility, and highlighting particular moral evaluations [5]. In ESA discourse, this often appears as competing portrayals of conservation versus economic cost, or federal protection versus local control. Through priming, repeated exposure to these frames conditions the criteria that audiences use when evaluating policies and political actors, influencing whether the ESA is judged in terms of ecological necessity, regulatory burden, or ethical obligation [6]. Empirical work shows that ESA coverage frequently emphasizes conflict and controversy, suggesting that media do not simply increase attention but also structure how the policy is interpreted and judged [2, 7, 8]. In this sense, media do not merely report ESA debates, they actively construct the communicative environment in which those debates unfold [9].

The importance of this narrative struggle becomes even clearer when compared with public support for the ESA. Survey evidence shows that support for the law has remained high and relatively stable over time, even as political debate around the ESA remains intense and polarized [?]. This creates an important puzzle: if public support is steady, why does ESA discourse remain so conflicted? One answer is that the policy debate is shaped not only by public

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preferences, but also by the narratives told by media, policymakers, and interest groups. Understanding how those narratives are formed, how they circulate, and how they influence interpretation is therefore essential to understanding endangered species politics itself.

Treating agenda-setting, framing, and priming as separate processes, however, can hide how policy communication actually works in practice. In domains like the ESA, these mechanisms are connected and operate together through broader narrative structures that link problems, actors, causes, and solutions into coherent stories. The Narrative Policy Framework (NPF) offers an integrative way to study this process by treating policy discourse as structured storytelling [?]. Within the NPF, setting directs attention, characters are assigned roles such as heroes, villains, and victims, plots establish causal relationships, and policy solutions express normative judgments about what should happen next. This makes the NPF especially useful for ESA debates, where species are often portrayed as victims, regulators as protectors or overreaching authorities, and policy choices as moral decisions about conservation and economic trade-offs [10?, 11]. Compared with approaches such as topic modeling, which capture word co-occurrence but not causal or evaluative meaning, the NPF offers a stronger way to understand how narratives structure policy conflict and communication [?].

This paper builds on that logic by asking how ESA narratives are constructed, how they move through media and political institutions, and whether they shape policy outcomes. This paper develops a prompt-engineered large language model framework for extracting ESA narratives using NPF-based coding. The goal is not only to measure narratives at scale, but also to test whether a theory-driven LLM approach can capture the structure of ESA discourse more effectively than traditional text-as-data methods. Taken together, the paper argues that ESA politics cannot be understood through law or science alone; it must also be understood as a communicative process in which narratives organize attention, interpretation, and evaluation, and ultimately shape how the law is debated and implemented [9?].

Large language models (LLMs) have significantly simplified many natural language processing tasks [12]. Although there are obvious advantages to this capability [13], it also provides tools that can influence public opinion [14]. Importantly, well-chosen adjustments can influence how readers understand a claim, how reliable a source seems, and how automated algorithms assess trustworthiness [15].

This paper investigates, for the first time, whether [describe your specific research question]. We consider [describe your approach]. The goal is [describe your goal].

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes our methodology. Section 4 presents the experimental evaluation. Section 5 concludes the paper.

2 Related Work

2.1 The Role of Media in Shaping Policy Attention, Interpretation, and Evaluation

The introduction to this dissertation argued that media institutions do not merely report on ESA debates but actively construct the communicative environment in which those debates unfold. This subsection traces the theoretical foundations behind that claim, situating ESA-specific media dynamics within the broader literature on how news coverage shapes political attention and judgment.

Emiliano and co-authors have debated the role of media in shaping public policy, particularly how media narratives influence public opinion and create political pressures that affect legislative decision-making [?]. Through their findings, they emphasize that one central pathway runs through agenda setting and the ways issues are framed for specific policies, such that media attention and narrative emphasis help determine which problems become politically urgent and how they are understood by elites and citizens. This perspective builds on classic agenda-setting theory, which argues that mass media do not simply mirror public concerns, but actively structure the public agenda by allocating scarce attention across competing issues [?]. When news organizations repeatedly highlight certain topics such as crime, inflation, or environmental crises, those topics become more salient in citizens' minds, shaping perceptions of what government should prioritize and, in turn, creating incentives for legislators to defend or oppose particular policy responses. Consistent with this view, studies show that sustained media coverage can elevate specific policy issues onto the political agenda, influence the salience of attitudes toward those issues, and thereby condition the space within which policy debates unfold [?].

Scholars have long debated the role of media in shaping public policy, particularly how media narratives influence public opinion and create political pressures that affect legislative decision-making [16]. One central pathway runs through agenda setting and the ways issues are framed for specific policies, such that media attention and narrative emphasis help determine which problems become politically urgent and how they are understood by elites and citizens alike. This perspective builds on classic agenda-setting theory, which holds that mass media do not simply mirror public concerns but actively structure the public agenda by allocating scarce attention across competing issues [4]. When news organizations repeatedly highlight certain topics — crime, inflation, environmental crises — those topics become more salient in citizens' minds, shaping perceptions of what government should prioritize and, in turn, creating incentives for legislators to defend or oppose particular policy responses. Sustained media coverage can elevate specific policy issues onto the political agenda, influence the salience of attitudes toward those issues, and condition the space within which policy debates unfold [4, 17].

This claim is not merely theoretical. (author?) [4]'s original study operationalized agenda setting empirically, comparing the frequency of issue coverage during the 1968 U.S. presidential election against survey responses identifying the “most important problem” facing the country, and found a strong correlation between media issue

salience and public issue importance — evidence that media attention shapes what citizens think *about* rather than what they think. Subsequent research has replicated and extended these findings across policy domains and political contexts [18, 19], though later studies show that agenda-setting effects are conditional rather than universal: they tend to be stronger for less noticeable or complex issues where citizens lack direct personal experience [20], and may weaken in fragmented media environments characterized by selective exposure [21]. Time-series analyses further suggest that the relationship between media and public salience may run in both directions, with public concern sometimes driving media coverage rather than the reverse [22] — a possibility directly relevant to this dissertation’s second chapter, which asks whether media lead congressional attention to the ESA, whether Congress leads media coverage, or whether the relationship is genuinely reciprocal.

Scholars have long debated the role of media in shaping public policy, particularly how media narratives influence public opinion and create political pressures that affect legislative decision-making [16]. One central pathway runs through agenda setting and the ways issues are framed for specific policies, such that media attention and narrative emphasis help determine which problems become politically urgent and how they are understood by elites and citizens alike. This perspective builds on classic agenda-setting theory, which holds that mass media do not simply mirror public concerns but actively structure the public agenda by allocating scarce attention across competing issues [4]. When news organizations repeatedly highlight certain topics — crime, inflation, environmental crises — those topics become more salient in citizens’ minds, shaping perceptions of what government should prioritize and, in turn, creating incentives for legislators to defend or oppose particular policy responses. Sustained media coverage can elevate specific policy issues onto the political agenda, influence the salience of attitudes toward those issues, and condition the space within which policy debates unfold [4, 17]. This claim is not merely theoretical. (author?) [4]’s original study operationalized agenda setting empirically, comparing the frequency of issue coverage during the 1968 U.S. presidential election against survey responses identifying the “most important problem” facing the country, and found a strong correlation between media issue salience and public issue importance — evidence that media attention shapes what citizens think about rather than what they think. Subsequent research has replicated and extended these findings across policy domains and political contexts [18, 19], though later studies show that agenda-setting effects are conditional rather than universal: they tend to be stronger for less noticeable or complex issues where citizens lack direct personal experience [20], and may weaken in fragmented media environments characterized by selective exposure [21]. Time-series analyses further suggest that the relationship between media and public salience may run in both directions, with public concern sometimes driving media coverage rather than the reverse [22] — a possibility directly relevant to this dissertation’s second chapter, which asks whether media lead congressional attention to the ESA, whether Congress leads media coverage, or whether the relationship is genuinely reciprocal.

Influencing issue salience, however, represents only one dimension

of media power. Media coverage may not only determine which issues receive attention but also shape how those issues are evaluated and interpreted, meaning agenda setting is only one of several mechanisms through which media can shape policy-relevant attitudes and behavior [23]. Framing and priming operate as two further, related but distinct mechanisms. Where agenda-setting research asks which issues become salient, framing research asks how alternative descriptions of the same issue change interpretation. Iyengar’s experiments on episodic versus thematic television news show that different frames of the same social problem shift whether citizens attribute responsibility to individuals or to society [24], extending earlier psychological work on gain-loss framing in decision-making [25]. Subsequent studies show that news stories foregrounding different core values — public order versus free speech, for instance — alter tolerance for controversial groups [26], and that framing effects are strongest when the frame originates from a credible source [27]. More recent reviews emphasize that framing is highly conditional, weakening when competing frames are present, when individuals hold strong prior attitudes, or as time passes without reinforcement [28, 29]. Framing, in other words, is best understood not as a uniform media effect but as a context-dependent mechanism operating under specific informational and audience conditions — a description that maps closely onto ESA discourse, where the same listing decision can be framed alternately as ecological necessity, regulatory overreach, or economic threat depending on the outlet and the audience being addressed.

Priming, the third mechanism, was originally investigated in connection with agenda setting, grounded in the idea that repeated media attention to an issue increases its accessibility in memory and, in turn, influences the criteria citizens use to evaluate political actors and policies [30]. (author?) [31] extend this logic, arguing that the more attention media gives to a particular issue domain, the more citizens are primed to incorporate what they know about that domain into broader political judgments — for instance, evaluations of presidential performance shifting depending on how heavily the economy has been covered. Yet priming evidence is mixed: a preregistered replication under contemporary media conditions finds no average priming effect at all [32], while a meta-analysis concludes that priming is strongest in political news contexts and among less knowledgeable audiences, weakening as individuals are exposed to more diverse media environments [33]. Together, these findings suggest that priming is a genuine but fragile mechanism, dependent on sustained issue emphasis and particular audience conditions rather than operating automatically whenever coverage increases.

These debates return to (author?) [34]’s classic observation that the media may not be successful in telling people what to think, but are remarkably successful in telling people what to think about. (author?) [35] sharpen this distinction by separating accessibility — the increased likelihood that certain considerations come to mind following repeated exposure — from applicability — the extent to which specific considerations are perceived as relevant to the evaluation at hand. In this framework, agenda setting and priming operate primarily through accessibility, while framing operates through applicability, shaping which considerations are seen as

appropriate or compelling for judging a given policy. This cognitive distinction offers a clearer map of how media attention ultimately feeds into political attitudes and policy-relevant judgments — but it also raises the central methodological tension this dissertation must confront: agenda setting, priming, and framing are conceptually distinguishable, yet because all three operate through overlapping processes of accessibility and applicability, disentangling them empirically is genuinely difficult. Determining whether an observed shift in ESA discourse reflects a change in issue salience, evaluative weighting, or interpretive framing requires not just careful theory but appropriately structured data and clearly distinguished outcome measures.

This precision matters directly for how this dissertation is designed. Agenda-setting analyses typically link time series of issue frequency in media to measures of issue salience in surveys or political agendas [4, 22]; priming studies examine whether exposure to an issue alters the weight that issue receives in political evaluations, typically through interaction models between media emphasis and issue-specific attitudes [31, 33]; and framing research holds the underlying problem constant while manipulating or identifying variation in the interpretive package attached to it, inferring effects from differences in attribution, value priorities, or policy support across conditions [24, 26, 27]. Simply counting how often the ESA is mentioned in the news or the Congressional Record would be sufficient for a first-cut agenda-setting analysis of the kind this dissertation's second chapter undertakes, but it would be inadequate on its own to identify priming or framing effects. Detecting priming would require modeling whether increased ESA coverage changes the degree to which economic, ecological, or partisan considerations predict legislators' evaluations or voting behavior; detecting framing would require identifying the problem definitions, causal attributions, and moral evaluations embedded within ESA discourse directly. Because all three mechanisms tend to rely on similar textual indicators — issue frequency, co-occurrence with evaluative language — their boundaries are often blurred in practice, which is precisely why this dissertation does not attempt to study them as three separate, narrowly-operationalized variables, but instead turns to a framework built to capture all three simultaneously within a single coherent narrative structure: the Narrative Policy Framework, introduced in the section that follows.

2.2 The Narrative Policy Framework

The previous subsection traced how agenda setting, framing, and priming operate as related but conceptually distinct media effects, and closed by noting that their boundaries are often blurred in practice because all three rely on overlapping textual indicators and overlapping cognitive processes of accessibility and applicability. This raises the question of whether the Narrative Policy Framework (NPF) offers unique analytical leverage over these alternative approaches, and how its core elements can generate testable hypotheses about policy dynamics in the ESA context that motivates this dissertation.

The NPF addresses this challenge by shifting analytical focus from

isolated media effects to integrated narrative structures that simultaneously organize attention, interpretation, and evaluation through four structural components: setting (which directs agenda attention), characters (which prime relevant actors), plot (which frames causal relationships), and policy solution (which prescribes normative evaluation) [36, 37]. Rather than treating agenda setting, priming, and framing as separable processes that must each be operationalized and tested on their own, the NPF maps its elements directly onto these cognitive mechanisms — setting to issue salience, characters to decision weights, plot to causal interpretation, and solution to evaluation criteria — while providing a computationally structured framework for text analysis that enforces narrative logic within observable discourse units [36, 38]. This line of thinking raises the possibility that narrative-oriented frameworks such as the NPF [39] may offer useful conceptual tools for examining how accessibility and applicability interact within policy communication, without presuming that any single media effect operates in isolation.

The NPF also offers distinct advantages over alternative theoretical frameworks that might otherwise be applied to ESA discourse. The Multiple Streams Framework captures problem-stream attention through policy windows but neglects the character-driven causality central to framing effects [40]; Policy Feedback Theory traces institutional effects on preferences over time but misses strategic elite storytelling [41]; and standard topic models identify thematic clusters without enforcing narrative coherence [42]. The NPF uniquely synthesizes agenda setting, priming, and framing into a single structure, treating policy communication as storytelling rather than as discrete shifts in issue salience or isolated frames [43]. Settings, characters, and causal plots specify not only which issues appear on the agenda, but also who is responsible for problems and which evaluative standards should apply, thereby integrating all three media-effects mechanisms into one coherent narrative structure [44]. (author?) [36] introduce the NPF specifically as a quantitative, structuralist, and positivist approach that acknowledges the importance of narratives in policy processes and seeks to enable more systematic and productive empirical debate over how stories influence public policy. In this sense, the NPF can be read as an attempt to reconcile and extend classic media-effects models rather than to replace them.

This narrative perspective has been demonstrated productively outside the environmental domain as well. (author?) [45] illustrates the analytical value of the NPF in the study of economic communication by combining a backward-looking conception of inflation narratives with large language models, showing how the NPF can be used to systematically analyze media narratives as causal explanations of economic outcomes. The same underlying structure — settings, characters, plots, and moral or policy conclusions — applies just as naturally to ESA discourse, where species are routinely cast as victims, regulatory agencies alternately as protectors or overreaching authorities, and policy choices as moral judgments about conservation and economic trade-off [10, 11, 46].

Despite these conceptual strengths, however, scholars have increasingly noted that the NPF has been more successful as a descriptive

framework than as a measurement framework. As several reviewers observe, most applications of the NPF have remained largely descriptive, relying on manual coding of a limited set of texts and offering few scalable tools for measuring narrative elements in large corpora [39]. This measurement gap is particularly acute when the research goal is to trace temporal feedback between media and policymaking venues, where narratives may originate in one arena and diffuse or transform as they move to another — precisely the question this dissertation’s second chapter takes up for ESA discourse between the press and Congress. A parallel observation arises in the study of economic narratives, where understanding behavior depends on the stories people tell, yet quantitative approaches to analyzing those narratives remain largely underdeveloped [47]. One of the central challenges identified in this literature is simply the difficulty of measuring narratives in written or spoken text [48]; any adequate measure must capture relationships between characters, concepts, organizations, or objects [49]. Traditional text-as-data methods in social science, which typically focus on word frequencies or broad topic patterns, fail to account for these relational structures, leaving the causal structure of narratives unmeasured [50, 51].

Although recent advances such as RELATIO and subsequent work on economic narratives were developed in macroeconomic contexts, the underlying methodological challenge is directly analogous in environmental and conservation policy. Central banks and finance ministries care how public narratives around inflation, growth, and debt shape expectations and behavior; legislators and agencies involved in ESA implementation must navigate narratives that link species protection to jobs, property rights, or media attention, and the emergence of dominant storylines also characterizes conflicts over endangered species. (author?) [48] address the measurement challenge directly by offering an open-source Python package, RELATIO, for measuring interpretable narratives from plain text documents, bridging the gap between conceptual frameworks like the NPF and scalable measurement and offering a promising path for turning descriptive insight into systematic, empirical analysis. Because media narratives evolve rapidly, (author?) [52] further emphasize that it is increasingly important not only to extract narratives from a given corpus but also to examine how they emerge and shift over time, extending the RELATIO pipeline by incorporating additional NLP models of different architectures. While these extensions improve the ability to track narrative dynamics, (author?) [45] note that traditional NLP-based approaches still face a fundamental limitation: although such models address scalability, they lack the integrated language comprehension required to fully capture the contextual meaning and relevance of narratives. To address this gap, (author?) [45] propose an integrated approach that leverages large language models such as GPT [53] to identify economic narratives directly from large text corpora.

This same limitation — scalable extraction without genuine language comprehension — defines the methodological gap this dissertation is built to close for the ESA. The emerging field of digital conservation text studies uses large-scale digital traces and textual corpora to track public interest and sentiment around biodiversity, sustainability, and conservation issues, and has expanded rapidly

over the past decade, yet this work has so far made only limited use of large language models for capturing how attitudes and narratives about conservation evolve over time, particularly in relation to policy debates and institutional decision-making. Building directly on the RELATIO and LLM-based economic narrative literature, this dissertation’s first empirical chapter develops a prompt-engineered large language model pipeline to extract NPF-structured narrative elements — settings, characters, plots, and morals — from ESA discourse at scale, evaluated against Structural Topic Modeling as a baseline rather than assumed superior outright. In doing so, it takes the NPF’s conceptual integration of agenda setting, priming, and framing developed earlier in this chapter and operationalizes it computationally, turning a framework that has so far been predominantly descriptive into one that can be applied systematically across the large, multi-venue corpora — news coverage and the Congressional Record — that ESA agenda-setting and diffusion questions require.

2.3 Computational and LLM-Based Approaches to Narrative and Policy Text

The previous subsection closed by noting that the Narrative Policy Framework has historically been more successful as a descriptive framework than as a measurement framework, with most applications relying on manual coding of a limited set of texts and offering few scalable tools for extracting narrative elements from large corpora [39]. This subsection traces how computational text analysis methods have attempted to close that gap, moving from general-purpose text-as-data methods through structural topic modeling and entity-relation extraction to the recent emergence of large language models, and shows where each step still falls short of the goal this dissertation pursues: extracting the full structure of an NPF narrative — setting, characters, causal plot, and moral or policy conclusion — simultaneously, rather than modeling text along a single dimension such as topic or sentiment.

The starting point for this line of work is the broader text-as-data tradition in political science, which demonstrated the promise and the limits of automated text analysis well before narrative-specific methods existed [50]. General text-as-data methods, including topic models and supervised classification, made it possible to analyze political text at a scale no team of human coders could match, but these methods were built to surface themes and word patterns rather than the relational structure — who did what to whom, and why — that defines a narrative. [?] extended this tradition with Structural Topic Modeling (STM), which models latent themes in open-ended text at scale while accounting for document-level covariates through metadata-informed prevalence estimation. STM represented a substantial advance in scalability and remains a natural baseline for narrative-adjacent text analysis, including in this dissertation’s first empirical chapter, but it identifies thematic clusters without enforcing any narrative coherence across them [42?]: a topic model can show that a corpus discusses, say, wolves, ranchers, and federal regulation in similar proportions, but it cannot say who is cast as the hero or villain of that story, nor what causal claim links those actors together.

Closing that relational gap required a different methodological move. (author?) [48] addressed the challenge of measuring narratives directly by developing RELATIO, an open-source Python package for measuring interpretable narratives from plain text through structured semantic relations between entities, applied originally to congressional speech. Unlike topic models, a RELATIO-style entity and relation extraction pipeline captures who is doing what to whom within a text, moving analysis closer to the causal and relational structure that NPF research requires [49]. (author?) [52] pushed this further by combining dynamic topic models with narrative-shift detection using temporal embedding alignment, allowing researchers to track how policy narratives shift over time in large text corpora rather than treating a corpus as a single static snapshot — a methodological step directly relevant to this dissertation’s interest in how ESA narratives move and transform between media and congressional venues over time.

Despite these advances, (author?) [45] identify a fundamental limitation that persists across this entire line of work: although dynamic topic models and entity-relation pipelines address scalability, they lack the integrated language comprehension required to fully capture the contextual meaning and relevance of a narrative as a whole. A relation-extraction model can recover that one entity acted on another, but it does not, on its own, understand whether that action constitutes a moral transgression, a policy solution, or an incidental detail — distinctions that are central to the NPF’s structural elements. To address this gap, (author?) [45] propose an integrated approach that leverages large language models such as GPT [53] to identify economic narratives directly from large text corpora, treating the LLM not merely as a faster classifier but as a tool capable of the kind of contextual reading that narrative extraction actually requires. This approach is further supported by [?], who benchmark GPT-style models systematically across natural language processing tasks and demonstrate that large language models exhibit remarkable proficiency on tasks requiring deep comprehension of language and context, underscoring their potential to capture complex narrative structures that traditional NLP pipelines struggle to extract.

Table 1 summarizes this progression and the specific gap this dissertation addresses. Each step in this literature — from general text-as-data methods, through structural topic modeling, through entity-relation extraction, to large language model benchmarking and economic narrative extraction — expands what can be measured computationally, but no prior study applies these tools to extract a complete NPF narrative structure from policy text in a single, explicit, repeatable procedure. Topic models capture themes but miss narrative structure entirely; recent LLM-based approaches demonstrate strong language comprehension but are not yet paired with an explicit methodological design for extracting NPF components specifically. This dissertation’s first empirical chapter closes that gap by developing a prompt-engineered LLM pipeline that treats prompt design itself as an algorithm, extracting all four NPF elements — setting, characters, causal plot, and policy or moral conclusion — simultaneously from each document in the ESA corpus,

and evaluating that pipeline directly against Structural Topic Modeling as a baseline rather than assuming the LLM-based approach is superior outright.

3 Methodology

3.1 Datasets

To investigate how narratives surrounding the U.S. Endangered Species Act (ESA) are constructed and evolve across different policy arenas, this study developed a multi-source text corpus consisting of news media articles and congressional documents. These two sources capture complementary dimensions of policy discourse: news media represent the narratives presented to the public, while congressional records document how legislators frame, debate, and justify ESA-related policy decisions. Together, they provide a comprehensive foundation for examining the interaction between media narratives, public discourse, and legislative deliberation.

The news media corpus was obtained through **ProQuest TDM Studio**, which provides full-text access to historical newspaper archives and other news publications. Articles were retrieved using a comprehensive set of ESA-related search terms, including *“Endangered Species Act”*, *“endangered species”*, *“threatened species”*, *“critical habitat”*, *“habitat conservation”*, *“U.S. Fish and Wildlife Service”*, *“National Marine Fisheries Service”*, and the names of numerous federally listed species. These search terms were designed to capture broad coverage of ESA-related reporting while minimizing unrelated environmental stories.

The resulting dataset contains article metadata—including publication date, publication title, publisher location, source type, authorship, article title, and full text—covering multiple decades of ESA reporting across national, regional, and local newspapers. This diversity allows the analysis to examine how narratives differ across media organizations and geographic regions while also tracking changes in framing over time.

Before analysis, several preprocessing steps were performed. Duplicate articles resulting from syndicated publications or repeated database entries were removed. Records lacking usable full text were excluded. Standard text cleaning procedures—including removal of formatting artifacts, HTML tags, excessive whitespace, and encoding inconsistencies—were applied to produce a consistent corpus suitable for natural language processing.

The second component of the dataset consists of congressional documents obtained from the **Congressional Record (CREC)**. These documents include floor speeches, debate transcripts, statements by members of Congress, extensions of remarks, committee-related discussions recorded in the Congressional Record, and other legislative communications that reference the ESA or federally protected species.

Congressional documents were collected using ESA-related keywords comparable to those used for the news corpus. The collection spans multiple congressional sessions, enabling analysis of legislative narratives over an extended period and across changing political contexts.

Unlike news articles, congressional records contain substantial amounts of procedural language, ceremonial statements, legislative scheduling information, and administrative text that are not directly

Table 1: Prior work on computational narrative and policy text analysis, and the gap this dissertation addresses.

Study	Goal	Methods used	Reference
Jones and McBeth (2010)	Establish the Narrative Policy Framework and define the four structural elements of narrative in policy discourse.	Conceptual NPF framework; no computational extraction.	(author?) [54]
Grimmer and Stewart (2013)	Demonstrate the promise and limits of automated text analysis for political science research.	General text-as-data methods including topic models and supervised classification.	(author?) [55]
Roberts et al. (2014)	Model latent themes in open-ended text at scale while accounting for document-level covariates.	Structural topic modeling (STM) with metadata-informed prevalence estimation.	(author?) [56]
Ash et al. (2024)	Measure interpretable narratives from plain text using structured semantic relations between entities.	RELATIO-style entity and relation extraction applied to congressional speech.	(author?) [57]
Shahriar et al. (2024)	Evaluate whether large language models can perform deep language understanding and context-sensitive classification tasks.	Systematic benchmarking of GPT-style models across natural language processing tasks.	(author?) [58]
Lange et al. (2025)	Track how policy narratives shift over time in large text corpora.	Dynamic topic models combined with narrative-shift detection using temporal embedding alignment.	(author?) [59]
Schmidt et al. (2025)	Identify economic narratives directly from large corpora without pre-specified category systems.	Integrated NLP pipeline combining entity extraction with large language model classification.	(author?) [45]

Note: LLM = large language model; STM = structural topic model; NPF = Narrative Policy Framework; NLP = natural language processing. The present study differs from all prior work in applying a structured five-layer prompt to extract setting, characters, causal plot, and policy moral simultaneously from each document, rather than modeling topics.

relevant to ESA policymaking. Consequently, an additional filtering stage was necessary to isolate substantive policy discussion.

This study employs a two-stage filtering process. First, paragraphs are segmented from the original congressional documents and evaluated using language-model perplexity scores to identify highly repetitive procedural text commonly found in legislative records. Second, a large language model (LLM) classifies each paragraph into one of three categories: **procedural**, **substantive ESA policy discussion**, or **mixed content**. Only paragraphs classified as substantive or mixed are retained for subsequent narrative analysis. This approach substantially reduces noise while preserving meaningful legislative discourse related to endangered species policy.

After preprocessing, both corpora are standardized into a common document format that includes document identifiers, publication or congressional dates, source information, and cleaned text. This standardized structure enables consistent downstream analyses using topic modeling, narrative extraction, and temporal trend analysis.

Maintaining separate media and congressional corpora also permits direct comparison of narrative evolution across institutional settings. Media texts reflect how ESA issues are framed for public audiences, whereas congressional texts reveal how policymakers construct, contest, and justify legislative positions. Comparing these sources provides insight into whether policy narratives emerging in the media correspond to those expressed within Congress and how competing narratives develop over time.

The resulting corpus is particularly well suited for studying narrative conflict surrounding the ESA because it captures discourse from both public communication and formal policymaking institutions. The combination of historical news reporting and legislative debate allows this study to examine not only what narratives emerge but also how they change over time, how they differ across

institutions, and how they contribute to broader policy debates concerning endangered species conservation.

3.2 Model Overview

We apply several techniques to [describe your approach]. Each step is designed to [describe goal] while preserving context and meaning. The general prompt used to instruct the LLM is as follows:

“You are an expert analyst that performs a single, deterministic transformation on input text. [TASK & RULES] Do not perform any other edits. Text: {text}. Only respond with the result, no explanation.”

Add here more text

3.3 Prompt-based Zero-Shot and Structured Narrative Extraction Approaches

Prompt-based zero-shot methods have become an important way to extract structured information from text without fine-tuning or task-specific training data. In this dissertation, the approach is not used for stance detection, but for ESA narrative extraction, where the goal is to identify the main policy event, code its narrative structure, and organize it into broader ESA policy categories. This makes the prompting pipeline a **structured extraction method** rather than a simple text-generation task. Zero-shot prompting is especially useful here because it allows the model to work directly from instructions and schema definitions, which reduces the need for manual annotation while keeping the analysis aligned with the research design [? ?].

The first stage of the pipeline is event identification, which asks the model to identify the single main ESA-related policy event discussed in a paragraph. This step is intentionally narrow: the prompt

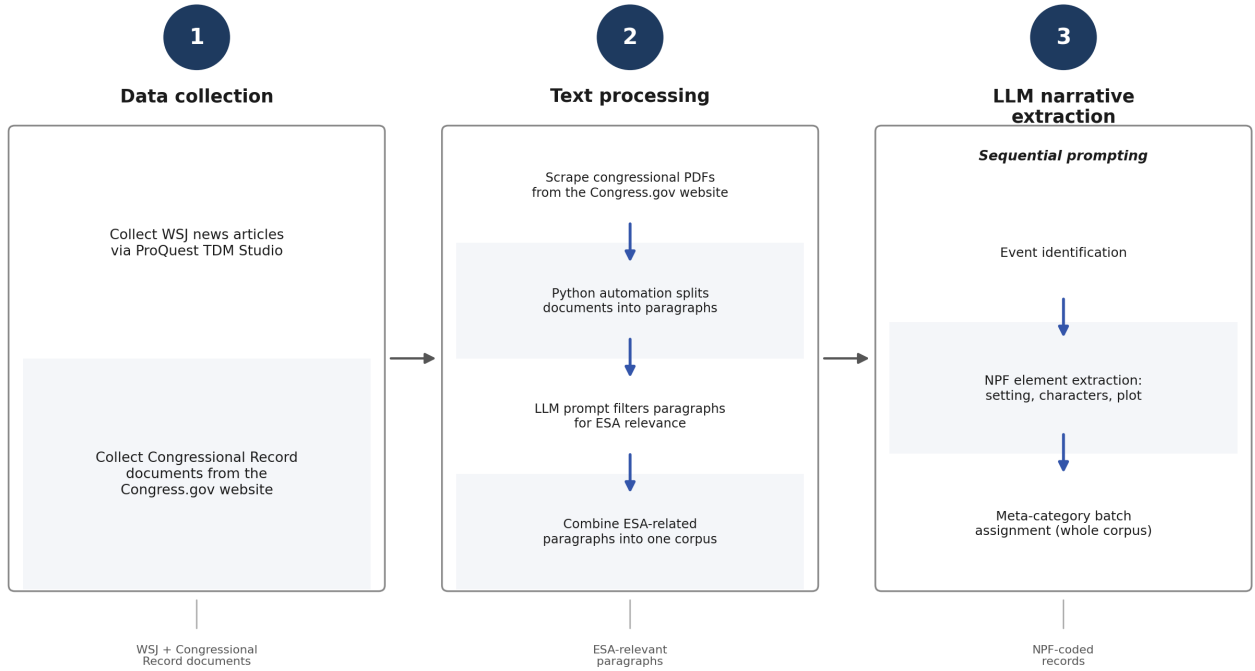


Figure 1: Overview of the three-stage research process for collecting, processing, and extracting NPF narrative elements from ESA policy text.

tells the model to focus on the policy event itself, not on the reporting form, actor names, or consequences. Such a design follows the logic of prompt-based zero-shot text classification, where the model is given a task description and a constrained output format so that it can generate a consistent label from unannotated text [? ?]. In my case, that means the model must return a short neutral event label, such as “species listing,” “delisting decision,” or “court ruling,” together with a brief description. This separation of event type from narrative interpretation is important because it gives the later NPF coding step a stable anchor.

The second stage uses the identified event as context for structured narrative extraction under the Narrative Policy Framework. Here, the prompt is designed to extract the setting, characters, plot, and moral from the same paragraph. This makes the task theory-driven rather than purely descriptive, since the model is not just summarizing text but identifying how the text constructs a policy narrative. Recent work on prompt-based structured extraction shows that complex information extraction tasks benefit from carefully designed prompt templates, staged workflows, and explicit output constraints, especially in zero-shot settings [? ?]. Your pipeline follows that same logic by decomposing narrative analysis into smaller, interpretable steps rather than asking the model to do everything at once.

The final stage groups the extracted event records into broader ESA policy meta-categories. This step is useful because it moves the analysis from paragraph-level coding to corpus-level patterning. Instead of grouping by text source or actor, the meta-category prompt groups by the type of ESA policy event, such as listing, delisting, habitat and recovery, litigation, legislative action, or agency rule-making. This kind of prompt-based classification is consistent with recent zero-shot text classification work that emphasizes the importance of explicit schemas, clear label spaces, and constrained output formats [? ?]. In this dissertation, the three prompts work together as a structured zero-shot pipeline that supports reproducible narrative coding while remaining aligned with NPF theory.

The variants generated across multiple scenarios lack human-annotated scores. To address this, we implemented a model to automatically detect [outcome], serving as a proxy for human judgment. Given the small dataset sizes, we avoided classical machine learning or LLM fine-tuning, instead used the following zero- or few-shot LLM strategies:

Naive prompting: This zero-shot approach relies solely on the LLM’s internal knowledge and reasoning, without supplying any additional contextual cues or examples.

In-Context Learning (ICL): This is a few-shot paradigm where an LLM receives a small set of labeled exemplars alongside the query [60]. Providing representative examples helps the model calibrate its decision boundary.

4 Experimental Evaluation

4.1 Code book

For this study, we used the following two publicly available datasets. **Dataset 1 [61]:** This dataset was collected through a large-scale survey. After preprocessing, the dataset contained [X] positive and [Y] negative items.

Dataset 2 [62]: In this dataset, respondents rated items on a five-point Likert scale. Averaged scores yielded [X] positive and [Y] negative items.

We used the [model name] model as the base, keeping it frozen in its pre-trained state. To fit within the model’s context window, each item was truncated to the first 350 tokens.

4.2 Proxy Model Selection

We tested the three proxy model strategies from Section 3, with accuracy results summarized in Table 2.

Table 2: F1 scores achieved by the considered proxy models.

Data	Strategy	F1 (wtd.)	F1 neg.	F1 pos.
3*Dataset 1	Naive prompt	51.02	19.57	54.88
	COLA	55.56	23.26	59.52
	ICL	69.36	17.24	75.76
3*Dataset 2	Naive prompt	39.32	18.87	42.67
	COLA	77.91	28.57	85.98
	ICL	74.55	38.71	80.41

4.3 Main Results

Table 3 shows the main results of our experiments. Across both datasets, multiple strategies consistently [describe findings]. Among these, [strategy X] yielded the highest performance ([value]%), demonstrating that [interpretation].

Table 3: Results by category and scenario on the two datasets.

Category	Scenario	Dataset 1	Dataset 2
Category A	Scenario 1	35.47	31.25
	Scenario 2	24.36	24.48
	Scenario 3	26.07	28.13
	Scenario 4	28.63	28.13
Category B	Scenario 5	26.07	34.38
	Scenario 6	40.60	62.50
Category C	Scenario 7	22.22	27.08
	Scenario 8	24.36	31.77
Category D	Scenario 9	22.65	31.77
Category E	Scenario 10	28.21	29.17

Overall, strategies such as [A], [B], and [C] consistently produced some of the highest scores across both datasets, highlighting the susceptibility of [outcome] to these subtle perturbations [63, 64].

5 Conclusions

This study demonstrates that [main finding]. Across two independent datasets and [N] principled scenarios, we observed that [specific result]. Limitations include the datasets’ size, evaluation of only one LLM, and a limited set of transformation scenarios.

References

- [1] U.S. Congress. Endangered species act of 1973, 1973. Public Law 93-205, 87 Stat. 884.
- [2] ? Hendrickson. Article on endangered species act coverage in major u.s. newspapers. *Journal/Source Unknown*, 2005.
- [3] ? Freedman et al. Article on public understanding of environmental issues through news media. *Journal/Source Unknown*, 2021.
- [4] Maxwell E. McCombs and Donald L. Shaw. The agenda-setting function of mass media. *Public Opinion Quarterly*, 36(2):176–187, 1972.
- [5] Robert M. Entman. Framing: Toward clarification of a fractured paradigm. *Journal of Communication*, 43(4):51–58, 1993.
- [6] Shanto Iyengar and Donald R. Kinder. *News That Matters: Television and American Opinion*. University of Chicago Press, 1987.
- [7] Lawrence C. Hamilton, Joel Hartter, and Andrew Bell. Framing wildlife policy in the news: How media narratives shape public perception of conservation. *Environmental Communication*, 14(2):234–249, 2020.
- [8] Diana C. Mutz. The influence of media on public opinion. *Annual Review of Political Science*, 18:47–66, 2015.
- [9] Robert Cox and Phaedra C. Pezzullo. *Environmental Communication and the Public Sphere*. SAGE, 2016.
- [10] Jeremy T. Bruskotter et al. Wildlife policy and political polarization. *Frontiers in Ecology and the Environment*, 21(2):94–, 2023.
- [11] Martin A. Nie. The endangered species act at twenty-five: A constructivist perspective on policy evolution. *Politics and the Life Sciences*, 2004.
- [12] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33:1877–1901, 2020.
- [13] Hunt Allcott and Matthew Gentzkow. Social media and fake news in the 2016 election. *Journal of Economic Perspectives*, 31(2):211–236, 2017.
- [14] Soroush Vosoughi, Deb Roy, and Sinan Aral. The spread of true and false news online. *Science*, 359(6380):1146–1151, 2018.
- [15] Carl I Hovland and Walter Weiss. The influence of source credibility on communication effectiveness. *Public Opinion Quarterly*, 15(4):635–650, 1951.
- [16] Matt Grossman. How media narratives shape policy and legislative pressure. *Journal of Political Communication*, 2022.
- [17] Maxwell E. McCombs. *Setting the Agenda: The Mass Media and Public Opinion*. Polity Press, 2nd edition, 2014.
- [18] Maxwell E. McCombs and Sebastian Valenzuela. Setting the agenda: Mass media and public opinion. *Polity*, 2020.
- [19] David H. Weaver. Thoughts on agenda setting, framing, and priming. *Journal of Communication*, 57(1):142–147, 2007.
- [20] Wayne Wanta and Salma I. Ghanem. Effects of agenda setting. *Mass Media Effects Research: Advances Through Meta-Analysis*, pages 37–51, 2007.
- [21] W. Lance Bennett and Shanto Iyengar. A new era of minimal effects? the changing foundations of political communication. *Journal of Communication*, 58(4):707–731, 2008.
- [22] Stuart N. Soroka. Issue attributes and agenda-setting by media, the public, and policymakers in canada. *International Journal of Public Opinion Research*, 14(3):264–285, 2002.
- [23] Dietram A. Scheufele and David Tewksbury. Framing, agenda setting, and priming: The evolution of three media effects models. *Journal of Communication*, 57(1):9–20, 2007.
- [24] Shanto Iyengar. *Is anyone responsible? how television frames political issues*. University of Chicago Press, 1994.
- [25] Amos Tversky and Daniel Kahneman. The framing of decisions and the psychology of choice. *Science*, 211(4481):453–458, 1981.
- [26] Thomas E. Nelson, Rosalee A. Clawson, and Zoe M. Oxley. Media framing of a civil liberties conflict and its effect on tolerance. *American Political Science Review*, 91(3):567–583, 1997.
- [27] James N. Druckman. On the limits of framing effects: Who can frame? *Journal of Politics*, 63(4):1041–1066, 2001.
- [28] Dennis Chong and James N. Druckman. A theory of framing and opinion formation in competitive elite environments. *Journal of Communication*, 57(1):99–118, 2007.
- [29] Sophie Lecheler and Claes H. de Vreese. Getting real: The duration of framing effects. *Journal of Communication*, 61(5):959–983, 2011.
- [30] Michael Schudson. The politics of narrative form: The emergence of news conventions in print and television. *Daedalus*, 111(4):97–112, 1988.

- [31] Jon A. Krosnick and Donald R. Kinder. Altering the foundations of support for the president through priming. *American Political Science Review*, 84(2):497–512, 1990.
- [32] Tetsuro Kobayashi et al. Priming effects in a contemporary media environment: A preregistered replication. *Political Communication*, 2017.
- [33] Sei-Hill Kim et al. Priming effects: A meta-analysis. *Communication Research*, 2018.
- [34] Bernard C. Cohen. *The Press and Foreign Policy*. Princeton University Press, 2015.
- [35] Vincent Price and David Tewksbury. News values and public opinion: A theoretical account of media priming and framing. *Progress in Communication Sciences*, 13:173–212, 1997.
- [36] Michael D. Jones and Mark K. McBeth. A narrative policy framework: Clear enough to be wrong? *Policy Studies Journal*, 38(2):329–353, 2010.
- [37] Michael D. Jones and Mark K. McBeth. A narrative policy framework: Clear enough to be wrong? *Policy Studies Journal*, 2010. See related discussion in companion article.
- [38] Elizabeth A. Shanahan, Michael D. Jones, and Mark K. McBeth. The narrative policy framework. In *Theories of the Policy Process*. Routledge, 2018.
- [39] Caroline Schlauffer et al. The narrative policy framework: A traveler’s guide to policy stories. *Policy Studies Journal*, 2022.
- [40] John W. Kingdon. *Agendas, Alternatives, and Public Policies*. Longman, updated 2nd edition, 2011.
- [41] Paul Pierson. When effect becomes cause: Policy feedback and political change. *World Politics*, 45(4):595–628, 1993.
- [42] David M. Blei, Andrew Y. Ng, and Michael I. Jordan. Latent dirichlet allocation. *Journal of Machine Learning Research*, 3:993–1022, 2003.
- [43] Michael D. Jones, Elizabeth A. Shanahan, and Mark K. McBeth. The science of stories: Applications of the narrative policy framework in public policy analysis. *Palgrave Macmillan*, 2014.
- [44] Elizabeth A. Shanahan, Michael D. Jones, Mark K. McBeth, and Claudio M. Radaelli. The narrative policy framework. In *Theories of the Policy Process*, pages 173–213. Routledge, 2018.
- [45] Tobias Schmidt, Kai-Robin Lange, Matthias Reccius, Henrik Müller, Michael Roos, and Carsten Jentsch. Identifying economic narratives in large text corpora – an integrated approach using large language models. *arXiv preprint arXiv:2506.15041*, 2025.
- [46] Jeremy T. Bruskotter et al. Tolerance for wildlife: A psychological perspective. *Frontiers in Ecology and the Environment*, 2018.
- [47] Robert J. Shiller. *Narrative Economics: How Stories Go Viral and Drive Major Economic Events*. Princeton University Press, 2019.
- [48] Elliott Ash et al. Relatio: Text semantics capture political and economic narratives. *Political Analysis*, 2024.
- [49] Steven A. Sloman and Aaron Sloman. Causal models: How people think about the world and its alternatives. *Oxford University Press*, 2009.
- [50] Justin Grimmer and Brandon M. Stewart. Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3):267–297, 2013.
- [51] Matthew Gentzkow, Bryan Kelly, and Matt Taddy. Text as data. *Journal of Economic Literature*, 57(3):535–574, 2019.
- [52] Kai-Robin Lange et al. Extending relatio for dynamic narrative tracking with modern nlp architectures. *Working Paper*, 2025.
- [53] Tom B. Brown et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, 2020.
- [54] Michael D. Jones and Mark K. McBeth. A narrative policy framework: Clear enough to be wrong? *Policy Studies Journal*, 38(2):329–353, 2010.
- [55] Justin Grimmer and Brandon M. Stewart. Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3):267–297, 2013.
- [56] Margaret E. Roberts, Brandon M. Stewart, and Dustin Tingley. Structural topic models for open-ended survey responses. *American Journal of Political Science*, 58(4):1064–1082, 2014.
- [57] Elliott Ash et al. Relatio: Text semantics capture political and economic narratives. *Political Analysis*, 2024.
- [58] Sakib Shahriar, Brady D. Lund, Nishith Reddy Mannuru, Muhammad Arbab Arshad, Kadhim Hayawi, Ravi Varma Kumar Bevara, Aashrith Mannuru, and Laiba Batool. Putting gpt-4o to the sword: A comprehensive evaluation of language, vision, speech, and multimodal proficiency. *Applied Sciences*, 14(17):7782, 2024.
- [59] Kai-Robin Lange et al. Extending relatio for dynamic narrative tracking with modern nlp architectures. *Working Paper*, 2025.
- [60] Qingxiu Dong, Lei Li, Damai Dai, Ce Zheng, Jingyuan Ma, Rui Li, Heming Xia, Jingjing Xu, Zhiyong Wu, Tianyu Liu, Baobao Chang, Xu Sun, Lei Li, and Zhifang Sui. A survey on in-context learning, 2024.
- [61] Mostofa Najmus Sakib, Md Shoaib Ahmed, Francesca Spezzano, and Anne Hamby. Understanding news consumers’ perceptions of believability: A study of real and fake news. *Association for Computing Machinery*, 2025.
- [62] Ömer Sümer, Efe Bozkir, Thomas Kübler, Sven Grüner, Sonja Utz, and Enkelejda Kasneci. FakeNewsPerception: An eye movement dataset on the perceived believability of news stories. *Data in Brief*, 35:106909, 2021.
- [63] Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration. *arXiv preprint arXiv:1904.09751*, 2019.
- [64] Daphne Ippolito, Daniel Duckworth, Chris Callison-Burch, and Douglas Eck. Automatic detection of generated text is easiest when humans are fooled. *arXiv preprint arXiv:1911.00650*, 2019.